

Sensor-Driven Landslide Detection System for the South Eastern Region of India

Suvankar Singha

School of Electronics Engineering.

KIIT Deemed to be University.

Bhubaneswar, Odisha

suvankarsinghan2107@gmail.com

Shankhosuvro Ghosh

School of Electronics Engineering.

KIIT Deemed to be University.

Bhubaneswar, Odisha

shankhosuvro.ghosh@gmail.com

Piyush Ranjan

School of Electronics Engineering.

KIIT Deemed to be University.

Bhubaneswar, Odisha

kashyap.piyush2003@gmail.com

Jaydeep Rath

KIIT International School.

Bhubaneswar, Odisha

Jaydeeprath2006@gmail.com

Sarita Nanda

School of Electronics Engineering.

KIIT Deemed to be University.

Bhubaneswar, Odisha

snandafet@kiit.ac.in

Abstract— Landslides, a pervasive natural calamity, can lead to catastrophic outcomes when they transpire in proximity to human settlements and infrastructural facilities. The identification and surveillance of landslides are paramount for mitigating the hazards associated with these geological phenomena. One innovative approach to landslide detection involves the utilization of Arduino-based sensor technology. These sensors are designed to monitor alterations in the terrain's gradient, serving as an indicator of potential landslide activity. A cost-effective early warning system can be engineered using an Arduino Uno board, a microcontroller predicated on the ATmega328P, in conjunction with an MPU 6050 accelerometer and gyroscope sensor, and a Soil and Humidity Sensor. This system can provide preliminary alerts for impending landslides, thereby enhancing the efficacy of evacuation procedures.

Keywords—*Landslide Detection, Arduino, Sensors ,accelerometer, gyroscope ,microcontroller*

I. INTRODUCTION

Landslides are a widespread natural disaster that can have catastrophic consequences when they occur near settlements and infrastructure. The detection and monitoring of landslides are essential for managing the risks associated with these events. Given the susceptibility of south-eastern regions to landslides, these monitoring systems play a crucial role in enhancing early warning capabilities and minimizing potential damages. Furthermore, our access to soil samples were from this region only. While the system can be generalized, it is imperative to account for these local factors to ensure its

effectiveness. Therefore, any attempt to deploy this system in a new region should involve a careful analysis of the local environmental conditions. Monitoring of landslip events using ALOS/PALSAR imaging based on the interpretation of geomorphological satellite data [1]. Geocube for landslip monitoring system that uses GPS sensors spaced out at least 5 kilometres apart can track the behaviour of landslip potential is also done by the state of the water content in the soil [2]. The resistivity approach, measurements of resistivity, water potential, and content of water are employed as parameters utilised in sensor prediction [3]. Fibre-optic has also been used to monitor landslides and spread over distance. Additionally, it [4] has a measuring resolution level of 1 metre Landslip early warning system employing microcontroller in sensor node to read data from vibration sensor and then transfer data to master node via WIFI [10] module [6]. Radar and optical remote sensing are used to map and monitor landslides [7]. A landslip monitoring system that uses a temperature sensor and an extensometer sensor can measure up to 1023 mm accurately [8].

The primary factors that are employed as indicators for landslides, which are the shifting of soil and ground water content, come from several research that have been undertaken. In general, geophysics and geology experts carried out landslip monitoring and mitigation system. The geoelectric technique was often utilised by researchers to examine the landslip tragedy, along with other techniques including smorphslope and geology

information system. Landslides typically occur on slopes due to a combination of factors, including rainfall levels, earthquake risk, and vibration from train traffic in a nearby location, which might result in a landslide when it crosses describes several initiatives to enhance the accuracy of landslip system detection.

The research focuses on evaluating the effectiveness of three sensors for landslide monitoring and mitigation systems: a slope sensor (MPU 6050), a soil moisture sensor, and an Arduino Uno R3 microcontroller. These sensors measure terrain inclination, moisture levels, and landslide risk. The Arduino Uno R3 microcontroller[9] serves as a central hub for data processing and communication, enhancing monitoring accuracy, scalability, and adaptability.

II. DESIGN AND STRUCTURE OF THE SYSTEM

A. Data Collection

One of the most crucial pieces of data derived from the IMU (Inertial Measurement Unit) sensor output is the pitch and roll angles. The body's snout and the flat surface create a pitch angle. It resembles the starting and landing positions of an aeroplane. By rolling the body left and right, a roll angle is created between the body and the flat surface. It resembles how an aeroplane rolls when it is in flight. The x- and y-axes are perpendicular to the gravity vector, therefore their values are both equal to zero. Only the z-axis will be impacted by gravity when the sensor body is on a flat surface. The x-axis value is no longer 0 when the body is pitched with angle Θ because the g vector is no longer perpendicular to it, but the g vector is still perpendicular to the y-axis. The Pitch angle between the body and the surface is equal to the angle between the g vector and the z-axis after pitching, according to geometry principles. The angle forms an angle opposite to that made by the g vector and the x-axis. Using the z- and x-axis values, we find the value of pitch:

$$\Theta = \tan^{-1}\left(\frac{a_x}{a_z}\right) \quad (1)$$

Similarly, we use the z-axis and y-axis to get the rolling angle:

$$\Phi = \tan^{-1}\left(\frac{a_y}{a_z}\right) \quad (2)$$

Utilizing the data collected, we can perform comparative analysis with predefined thresholds, thereby enabling the system to issue a warning if either the Pitch angle or Tilt angle exceeds the established threshold values. This proactive approach to monitoring and threshold-based alerting ensures the timely identification of deviations in pitch and tilt angles, contributing to enhanced safety and risk mitigation measures. For more understanding, please refer to Fig 1.

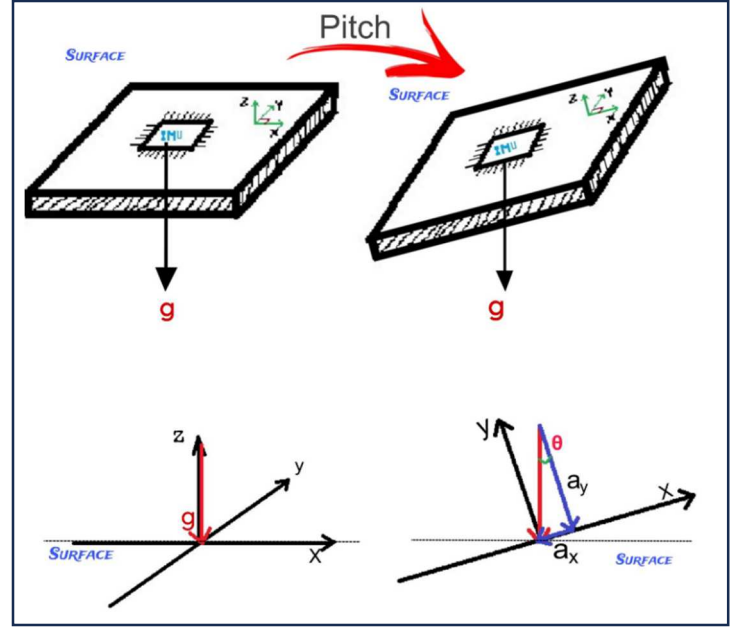


Fig 1- An Illustration for the Θ angle showing how it is complementary to the angle formed between the g vector and the x-axis.

B. Display

Using Microsoft Excel, we successfully visualized incoming serial data over USB on an X/Y axis. The X-axis, representing time, and the Y-axis, representing sensor data values, are both auto-scaled for convenience. The Arduino board plays a pivotal role in processing raw data obtained from sensors. This processing may involve crucial tasks such as converting raw values into more interpretable units, like degrees per second for gyroscope data or g's for accelerometer data. Additionally, noise filtering and amalgamation of data from multiple sensors contribute to refining the accuracy of the sensor readings. Following this preprocessing, the Arduino board transmits the processed sensor data to Microsoft Excel via serial communication. Each set of sensor readings, comprising accelerometer and gyroscope data for each axis, is meticulously plotted on separate lines within Excel. The resulting plots, namely Fig. 5(a), Fig. 5(b), Fig. 5(c), have been derived from the datasets meticulously presented in TABLE 2, TABLE 3, and TABLE 4. These graphical representations offer a dynamic and informative insight into how sensor readings evolve over time. Such visualizations prove instrumental in comprehending sensor behavior, identifying and rectifying potential issues during data acquisition, and facilitating detailed analysis of motion patterns. With the ability to delve into individual axes, these plots enable a nuanced examination of sensor performance and contribute significantly to the development, debugging, and optimization of sensor-based applications.

III. Operational approach for landslide detection

1. **Gyroscope and Accelerometer:** The MPU6050 sensor module is a space-efficient package that includes a 3-axis accelerometer, gyroscope, and temperature sensor. It can be connected to microcontrollers like Arduino Uno R3 using I2C communication protocol. The connection is established using four pins: GND, VCC, SCL, and SDA. The MPU6050 sensor has a Digital Motion Processor and a FIFO buffer, which collects sensor data and performs calculations. The Arduino is signalled with an interrupt, which triggers the Arduino's interrupt routine, checking the FIFO buffer's data.

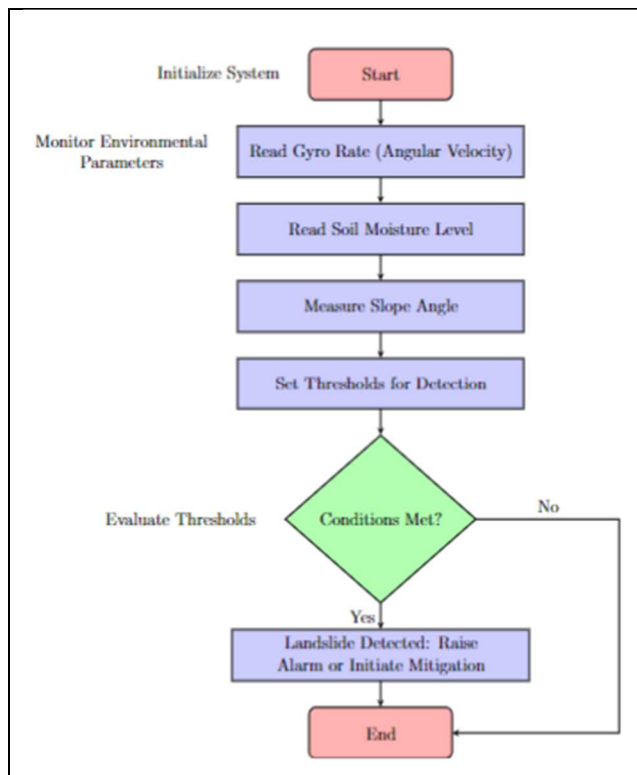


Fig. 2 Flowchart

2. **Soil Moisture Sensor:** The Soil Moisture Sensor measures soil moisture content and can be connected to microcontrollers like Arduino Uno R3 for applications like automated irrigation systems and digital agriculture. It uses electrical conductivity to convert soil conductivity into an analog value, which can be read by the Arduino's analog-to-digital converter (ADC).

The data from each sensor is continuously acquired and processed by the Arduino. The threshold values for each sensor are compared to the sensor readings, and if the sensor reading exceeds the threshold value, it is considered a potential landslide detection. To get an idea of how our Landslide Detection model works please refer Fig. 1 and to understand how the model works please refer to the Fig. 2, where the basic idea/working of the model is shown.

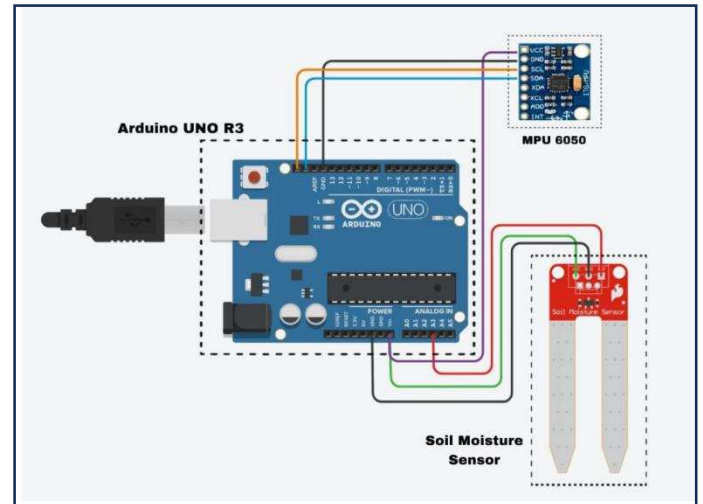


Fig. 3 Circuit Diagram

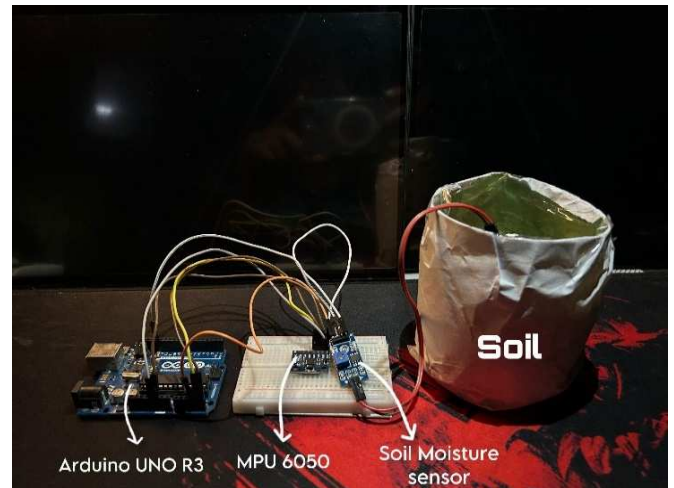


Fig. 4 Hardware Model

TABLE 1. Comparison of work done in previous years

Author	Work	Methodology	Advantages	Disadvantages
Schlögel R et al.	Geomorphology Monitoring the deformation of landslides using ALOS and PALSAR images.[1]	Provides a geomorphologically-based approach for deciphering L Band ALOS/PALSAR interferograms produced by DSAR	• Use of PALSAR techniques • Use of C Band and X band satellite SAR sensors	• Cost is more • MPU 6050 and Moisture sensor has not been used
Benoit et al.	Geocube's wireless, inexpensive GPS network is used to track landslide displacements.[2]	Presents the Geocube system and the findings of a field campaign using a wireless, low-cost network.	• Use of GPS • Use of Geocube	• Network Delay
Crawford MM et al.	Using electrical measurements in the field, assess active landslides. [3]	Presents how electrical resistivity can be used as a tool for detecting landslides.	• Use of water content reflectometers • Use of Water Potential sensors	• MPU 6050 has not been used
Yu Z et al.	Monitoring of landslides using an optical high-resolution distributed strain sensing system. [4]	Presents a Rayleigh backscatter based coherent optical time domain reflectometry technique to realize long-distance, real-time monitoring.	• Long Distance • Real time monitoring	• Readings not as accurate as there is no slope calculation and Tilt angle.
Bovenga et al.	Employing a corner reflector and satellite SAR interferometry to detect landslides to reduce risk. [5]	Presents findings from integrating high resolution TerraSARX (TSX) data with Multi Temporal InSAR algorithms for monitoring Carlantino landslip.	• Use of Corner Reflectors • Use of SAR techniques	• Cost is more. • MPU 6050 and humidity sensor is not used.
] Biansoongnem S et al.	Creation of a Low Cost Vibration Sensor Network for a Landslide Early Warning System . [6]	This study describes a low-cost WIFI sensor network-based early warning system for landslides utilising the microcontrollers PIC12F683 and ATmega328.	• Use of WIFI sensor • Use of Vibration Sensor	• MPU 6050 and soil moisture sensor is not used. • Only vibration sensor is used which is not efficient.
Casagli N et al.	Environment and Society employing radar and optical remote sensing to map and monitor landslides. [7]	This study focuses on the SAFER (Services and Applications For Emergency Response) project's Landslip Thematic services for inventory mapping, monitoring, and quick mapping utilising Earth observation.	• Use of Radar and optical remote sensing • Use of SAR techniques	• MPU 6050 and soil moisture sensor is not used. • Not Real-time.
Guerriero L et al.	A cheap wire extensometer powered by Arduino ® for monitoring earth flow [8]	Presents a low cost Arduino based wire extensometer which incorporates the use of Precision Position Transducer and Temperature sensor	• Use of Precision position Transducer • Use of Temperature sensor	• MPU 6050 and Soil Moisture sensor is not used. • Cost is more.

IV. RESULTS

TABLE 2. Data collected from the prototype during no soil movement and rainfall (Optimal Condition)

Data collected from the prototype during no soil movement and rainfall (Optimal Condition)									
<i>X-axis (Accl)</i>	<i>Y-axis (Accl)</i>	<i>Z-axis (Accl)</i>	<i>X-axis (Gyro)</i>	<i>Y-axis (Gyro)</i>	<i>Z- axis(Gyro)</i>	<i>X-axis (Angle)</i>	<i>Y-axis (Angle)</i>	<i>Z-axis (Angle)</i>	<i>Soil Moisture (%)</i>
0.09	-0.14	8.65	-0.12	0.05	-0.01	-0.92	0.62	146.08	6.84
0.09	-0.15	8.63	-0.12	0.05	-0.01	-0.97	0.62	147.41	6.84
0.07	-0.16	8.56	-0.12	0.05	-0.01	-1.07	0.5	155.17	6.94
0.11	-0.11	8.64	-0.11	0.05	-0.01	-0.73	0.73	135	6.84
0.07	-0.07	8.58	-0.12	0.05	-0.01	-0.5	0.46	136.91	6.34
0.04	-0.1	8.47	-0.12	0.05	-0.01	-0.66	0.29	156.3	6.94
0.08	-0.11	8.56	-0.12	0.05	-0.01	-0.72	0.51	144.58	6.74
0.03	-0.08	8.59	-0.12	0.05	-0.02	-0.54	0.21	159.08	6.94
0.04	-0.12	8.58	-0.12	0.05	-0.01	-0.8	0.29	160.2	6.94
0.14	-0.17	8.64	-0.12	0.04	-0.02	-1.14	0.9	141.63	6.54

TABLE 3. Data collected from the prototype during soil movement (Landslide may occur)

Data collected from the prototype during soil movement (Landslide may occur)									
<i>X- axis(Accl)</i>	<i>Y- axis(Accl)</i>	<i>Z- axis(Accl)</i>	<i>X- axis(Gyro)</i>	<i>Y-axis(Gyro)</i>	<i>Z- axis(Gyro)</i>	<i>X-axis(Angle)</i>	<i>Y-axis(Angle)</i>	<i>Z-axis(Angle)</i>	<i>Soil Moisture %</i>
1.07	9.58	-0.08	-3.57	-0.84	-0.37	90.46	94.1	6.36	22.07
0.5	0.28	8.58	-0.12	0.04	-0.01	1.85	3.32	60.85	23.96
0.52	0.24	8.65	-0.12	0.05	-0.01	1.59	3.47	65.46	23.56
0.46	0.2	8.6	-0.12	0.05	-0.01	1.32	3.04	66.51	23.88
0.41	0.6	8.77	0.04	-0.01	-0.19	3.93	2.69	34.32	23.46
0.15	0.15	8.55	-0.13	0.05	-0.02	1.03	0.98	43.63	22.67
0.08	-0.19	8.71	-0.13	0.04	-0.02	-1.26	0.52	157.58	22.56
0.33	-0.41	8.52	-0.11	0.08	-0.02	-2.77	2.22	141.26	22.11
0.15	-0.34	8.69	-0.13	0.03	-0.01	-2.26	0.96	156.9	21.54
-0.08	-0.14	8.63	-0.13	0.03	-0.01	-0.95	-0.54	-150.46	23.56
0.17	-0.37	8.66	-0.12	0.05	-0.01	-2.42	1.11	155.42	23.98

TABLE 4. Data collected from the prototype during soil movement and rainfall (Possibility of landslide increases)

Data collected from the prototype during soil movement and rainfall (Possibility of landslide increases)									
<i>X- axis(Accl)</i>	<i>Y- axis(Accl)</i>	<i>Z- axis(Accl)</i>	<i>X-axis(Gyro)</i>	<i>Y-axis(Gyro)</i>	<i>Z-axis(Gyro)</i>	<i>X-axis(Angle)</i>	<i>Y-axis(Angle)</i>	<i>Z-axis(Angle)</i>	<i>Soil Moisture%</i>
1.07	9.58	-0.08	-3.57	-0.84	-0.37	90.46	94.1	6.36	72.87
0.5	0.28	8.58	-0.12	0.04	-0.01	1.85	3.32	60.85	73.46
0.52	0.24	8.65	-0.12	0.05	-0.01	1.59	3.47	65.46	73.46
0.46	0.2	8.6	-0.12	0.05	-0.01	1.32	3.04	66.51	73.46
0.41	0.6	8.77	0.04	-0.01	-0.19	3.93	2.69	34.32	73.46
0.15	0.15	8.55	-0.13	0.05	-0.02	1.03	0.98	43.63	82.67
0.08	-0.19	8.71	-0.13	0.04	-0.02	-1.26	0.52	157.58	82.51
0.33	-0.41	8.52	-0.11	0.08	-0.02	-2.77	2.22	141.26	62.11
0.15	-0.34	8.69	-0.13	0.03	-0.01	-2.26	0.96	156.9	61.88
-0.08	-0.14	8.63	-0.13	0.03	-0.01	-0.95	-0.54	-150.46	53.56
0.17	-0.37	8.66	-0.12	0.05	-0.01	-2.42	1.11	155.42	63.87
0.25	-0.36	8.68	-0.12	0.04	-0.02	-2.4	1.66	145.36	22.34

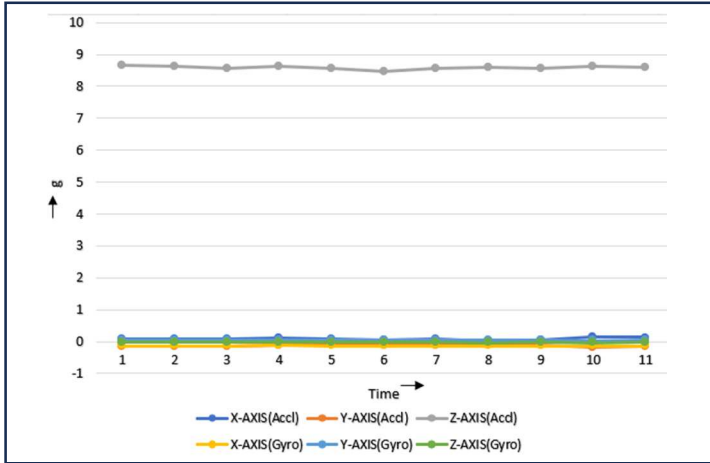


Fig. 5(a) Data collected from the prototype during No movement and rainfall (Optimal Condition)

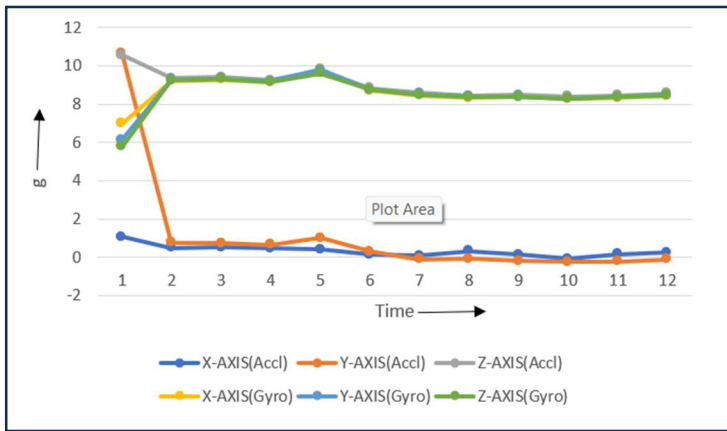


Fig. 5(b) Data collected from the prototype during soil movement (Landslide may occur)

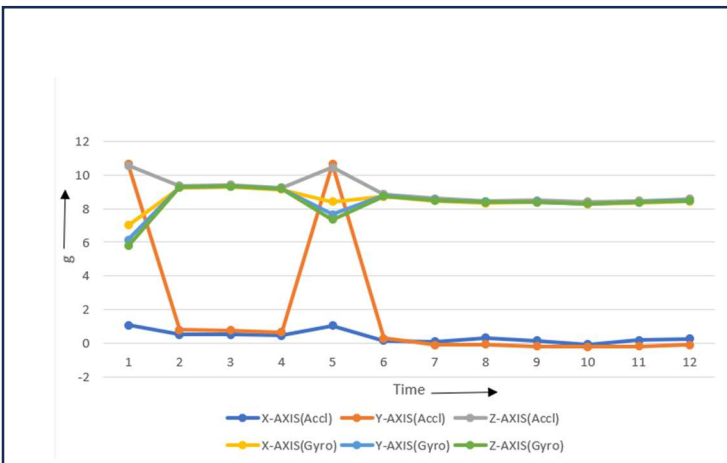


Fig. 5(c) Data collected from the prototype during soil movement and rainfall (Possibility of landslide increases)

V. DISCUSSION OF RESULTS AND CONCLUSION

The depicted figures (5a, 5b, and 5c) provide a comprehensive representation of the data derived from the prototype across various conditions. The first set captures the scenario of no movement and concurrent rainfall, portraying an optimal condition for analysis. The second set delves into the realm of soil movement, signifying a potential landslide event. Lastly, the third set encapsulates the combined effects of soil displacement and rainfall, shedding light on an elevated probability of a landslide occurrence. These visualizations serve as invaluable insights into the prototype's performance under diverse circumstances, offering a nuanced understanding of the interplay between soil dynamics and meteorological factors. By scrutinizing the intricate details presented in each figure, researchers and stakeholders can glean critical information regarding the system's responsiveness to environmental variables and its capacity to discern varying levels of landslide susceptibility. This comprehensive approach underscores the significance of the collected data in enhancing our comprehension of landslide dynamics and fortifying early warning systems.

In general, these results emphasize the prototype's ability to detect variations in soil conditions and its responsiveness to the combined influence of soil movement and rainfall. The findings underscore the prototype's potential as a tool for early landslide detection and prevention, providing valuable information for mitigating the risks associated with unfavorable environmental conditions. Further analysis and refinement of the prototype could contribute to enhancing its effectiveness in landslide prediction and risk management.

VI. REFERENCES

- [1] Schlögel R, Doubre C, Malet J, Masson F. Geomorphology Landslide deformation monitoring with ALOS / PALSAR imagery : A D-InSAR geomorphological interpretation method. *Geomorphology* [Internet]. Elsevier B.V.; 2015; 231:314–30.
- [2] Benoit L, Briole P, Martin O, Thom C, Malet J, Ulrich P. Monitoring landslide displacements with the Geocube wireless network of low-cost GPS. *Eng Geol* [Internet]. Elsevier B.V.; 2015;195:111
- [3] Crawford MM, Bryson LS. Assessment of active landslides using field electrical measurements. *Eng Geol* [Internet]. Elsevier; 2018;233(June 2017):146–59.
- [4] Yu Z, Dai H, Zhang Q, Zhang M, Liu L. Optic High-resolution distributed strain sensing system for landslide monitoring. *Opt - Int J Light Electron Opt* [Internet]. Elsevier GmbH; 2018;158:91-6
- [5] Bovenga F, Pasquariello G, Pellicani R, Re A, Spilotro G. Catena Landslide monitoring for risk mitigation by using corner reflector and satellite SAR interferometry : The large landslide of Carlantino (Italy). 2017;151:49–62.
- [6] Biancoognern S, Plungkang B, Susuk S. Development of Low Cost Vibration Sensor Network for Early Warning System of Landslides. *Energy Procedia* [Internet]. The Author(s); 2016;89:417–20.
- [7] Casagli, N., et al. "Landslide mapping and monitoring by using radar and optical remote sensing: Examples from the EC-FP7 project SAFER." *Remote sensing applications: society and environment* 4 (2016): 92-108.
- [8] Guerriero L, Guerriero G, Grelle G, Guadagno FM, Revellino P. Brief Communication : A low cost Arduino ® -based wire extensometer for earth flow monitoring. 2017;(iii):881–5.
- [9] Dasgupta, Anubhav Ranjan, et al. "Smart-Home Automation using AI Assistant and IoT." 2021 4th International Conference on Recent Trends in Computer Science and Technology (ICRTCTST). IEEE, 2022.
- [10] Mukherjee, Dibyarup, et al. "Real Time Agricultural Monitoring with Deep Learning Using Wireless Sensor Framework." 2023 International Conference on Electrical, Electronics, Communication and Computers (ELEXCOM). IEEE, 2023.